

Problem

Through derivation, express the z parameters in terms of the **ABCD** parameters.

Step-by-step solution

Step 1 of 2

The transmission parameter equations are

$$V_1 = AV_2 + B(-I_2)$$

$$I_1 = CV_2 + D(-I_2)$$

$$\text{or, } V_2 = \left(\frac{1}{C}\right)I_1 + \left(\frac{D}{C}\right)I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Step 2 of 2

Again

$$V_1 = AV_2 - BI_2 = A\left[\left(\frac{1}{C}\right)I_1 + \left(\frac{D}{C}\right)I_2\right] - BI_2$$

$$\left(\frac{A}{C}\right)I_1 + \left(\frac{AD+BC}{C}\right)I_2$$

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

So, we have

$$Z_{11} = \frac{A}{C}, \quad Z_{12} = \frac{AD+BC}{C}, \quad Z_{21} = \frac{1}{C} \quad \text{and} \quad Z_{22} = \frac{D}{C}$$